

Design Viability Judgment

Our proposed solution, a mechanical pressure gauge integrated with a 3D-printed body, is designed to meet our high-priority requirements (durability, accuracy, simplicity/replicability, affordability, and portability). Below, we provide credible evidence supporting the design's viability through a cost analysis, proof of concept, and a risk assessment with mitigation strategies.

1. Cost Analysis (Affordability)

One of the primary goals of our design is to ensure affordability and timeliness so that 10–20 units can be produced for classroom use within a reasonable time. Below is a breakdown of the estimated costs per unit:

Component	Estimated Cost Per Unit	Source	Date Bought
3D Printed Tube, Screw, Mouthpiece	\$1.00	Group Member Owned Printer	1/15/25
Vacuum Gauge with Steel Case	\$14.40	McMaster Carr	2/11/2025
Compression Spring	\$8.09	McMaster Carr	2/11/2025

This total cost of \$23.49 per unit is within the budget and we can continue to look for other sources to make it more cost effective. Moreover, the mechanical nature of the device eliminates the need for electronics.

2. Proof of Concept (Accuracy and Functionality)

To test the viability of the pressure measurement system, we used a commercially available mechanical pressure gauge and manually created a preliminary 3D-printed casing. The gauge reliably displayed pressure values within ± 1 cm H₂O when compared to a standard laboratory-grade manometer, meeting the required accuracy threshold.

The simplicity of using a mechanical pressure gauge ensures reliability and eliminates potential errors from electronic components, making the device more robust for repeated use in a classroom setting.

3. Risk Assessment and Mitigation Plan

A risk assessment was performed to identify potential risks associated with the design, along with mitigation strategies:

Component/Sub system	Risk	Mitigation Strategy	Justification (Evidence)
Mechanical pressure gauge	Inaccuracy over time due to wear	Select a high-quality, medical-grade gauge	Supported by supplier specifications
3D printed body	Structural failure	Use durable PLA+ filament for printing	Expert recommendation on filament type for strength
Tubing and mouthpiece	Hygiene concerns or material degradation	Use disposable, replaceable mouthpieces	Standard practice in medical respiratory devices

Assembly	Inconsistent builds among units	Create a standardized assembly guide	Ensures consistent and replicable results
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By mitigating these risks, we increase confidence in the device’s ability to meet the durability and accuracy requirements over an extended period of use.

4. Expert Feedback (Credible Evidence)

Two experts, Dr. Joseph Sarver and Jaimie Dougherty, reviewed our initial design. Both confirmed that the approach of using a mechanical gauge housed in a 3D-printed body is feasible and viable for the intended classroom application.

- Dr. Sarver emphasized the importance of selecting a durable pressure gauge to ensure long-term accuracy.
- Ms. Dougherty provided feedback on improving the portability of the device, which influenced our decision to keep the design compact and lightweight.

Their feedback strengthens our confidence in the design’s viability, and we have incorporated their suggestions into our final design.

5. Conclusion

Our design solution, consisting of a mechanical pressure gauge housed in a durable 3D-printed body, is both realistic and viable. The affordability is supported by a detailed cost analysis, the accuracy has been validated through a proof of concept, and potential risks have been mitigated through thoughtful design choices.

With expert feedback confirming the feasibility of our approach and our risk mitigation strategies in place, we are confident that our device will successfully meet all high-priority design requirements in a cost-effective manner.